Spice Route: an AI waiter for restaurant ordering and table booking

Product Requirements Document

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Version	v1 MVP
Date	3 November 2026
Status	Live and in use as a working product
Live product	restaurant-bot-three.vercel.app
Source code	github.com/djtrident1998/Restaurant-Bot

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1. Summary

Spice Route is an AI waiter for a restaurant. A diner can ask questions about the menu ("something vegetarian, mild, under ₹300"), build a cart by chat or voice, place an order, book a table, cancel, and split the bill with friends.

I built it around one product idea: **in food ordering, trust matters more than fluency.** A chatbot that sounds smart but invents a dish, quotes a wrong price or mishandles an allergy creates a wrong order, a safety problem or a refund dispute. So the AI in this product is allowed to talk, and nothing else. Everything that involves money, safety or consent is done by code that the AI cannot override:

- The AI can only get dishes, prices and allergens by calling a menu lookup. It never sees the menu in its prompt, so it has nothing to invent from.
- Totals, refunds, bill splits and table assignments are calculated by code, never by the AI.
- An order is placed only when the customer presses a confirm button. The server records that press. A typed "yes" is not enough, and the AI has no tool to place, cancel or refund anything.
- A saved allergy profile (the "Allergen Shield") flags every dish in code.
- **A chain of fallback models keeps the product running and keeps the cost down.** Free models answer first. If one is rate-limited or down, the next one takes over without the customer noticing, and the paid model is the last resort behind a hard daily cap. Most steps (menu, cart, checkout, cancel, booking) need no AI call at all, so they keep working even if every model is unavailable.

On my 46-case evaluation set, the tuned version made no invented-dish or invented-price errors, against a 4.3% hallucination rate for the first prompt. I treat that as an early signal, not proof: it was a single run on a set I wrote myself (details in section 9). 759 automated checks cover the rules that must never break.

This document describes the product as it stands, explains the decisions behind it, and sets out how I would measure it, evaluate the AI, and take it from here.

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2. Context and problem

2.1 Why this matters now

Conversational ordering is arriving in food delivery. In January 2026 Swiggy let customers order food, groceries and tables through ChatGPT, Claude and Gemini. Zomato took the other route and built an assistant inside its own app (launched September 2023). Both are aggregators. An individual restaurant has no equivalent for its own menu and its own bookings.

The risk is also public. McDonald's ended its AI drive-thru test at more than 100 US restaurants in 2024 after order errors went viral. Vendors who sell AI ordering tools say the same thing in their own guidance: answers about allergens must come from verified menu data and never from a model's general knowledge. A fluent assistant that is occasionally wrong is not good enough here.

2.2 The problem

Diners ask the same questions again and again: is it vegetarian, is it mild, what is under ₹300, does it contain nuts, do you have X. Menu PDFs cannot answer questions. General-purpose chatbots can, but they invent dishes and prices. In ordering, an invented fact turns into a wrong order, an allergen incident or a refund.

Problem statement: diners want quick, conversational help to order and book, but an assistant that can state a wrong dish, price, allergen or total, or act without the customer's consent, is not safe to put in front of them.

2.3 What I have and have not validated

This problem statement comes from desk research and my own understanding of how restaurant ordering works. I have not yet interviewed diners or restaurant owners, and I could not find verifiable public data on how aggregator AI assistants perform (conversion, error rates, retention). The plan to close this gap is in section 12. Everything below should be read with that in mind.

2.4 Who feels the problem

Who	Pain	How the product responds
Restaurant operator	Staff time goes on repeated questions; wrong orders cost money and trust	Grounded answers from the real menu at any hour
Restaurant operator	Fear of an AI taking actions it shouldn't	The AI has no tool to order, cancel or refund
Restaurant operator	Unpredictable AI bills if traffic spikes	Free models first, hard cap on the paid one, no-AI paths
Restaurant operator	Empty tables and booking friction	Live slots, no double-booking, pre-order once a table is booked
Diner	Allergy anxiety and unclear menus	Dish-level flags from a saved profile
Diner	Awkward group bills	Split that always adds up to the rupee

3. Users and jobs to be done

User	Priority	What they are trying to do
Diner with a dietary need (vegetarian, allergy, spice limit)	Primary	See what is safe for them, quickly
Decisive diner	Primary	Build a cart, see the exact total, confirm
Group host	Secondary	Order for a table and split fairly
Returning customer	Secondary	Re-check an order, rate it, get an invoice, cancel
Independent restaurant owner or manager (1 to 20 outlets, India)	Future buyer	Answer common questions and fill tables with less staff time

The jobs, in the customer's words:

- "I have a nut allergy. Show me what I can eat."
- "Add two paneer tikka and three naan. What's my total?"
- "Do you have sushi?" (they expect an honest no)
- "I want a table on Saturday for four, then I'll pre-order."
- "I changed my mind. Cancel it."
- "Split this between six of us."
- "Show me what you store about me, or delete it."

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4. Goals, non-goals and principles

4.1 Goals

1. **Trust:** no invented dishes, prices or totals. Allergen information comes only from menu data.
2. **Control:** no irreversible action (order, cancel, refund, booking) without clear intent recorded by the server.
3. **Speed:** from question to confirmed order in a few taps, and a booking in under a minute.
4. **Affordability:** survive a sudden traffic spike without a runaway AI bill.
5. **Credibility:** a sound baseline for accessibility, privacy and security, with the proof (tests, evals, guardrail write-up) visible.

4.2 Non-goals for this version

I kept the scope tight so the trust questions could be answered properly first.

- Payment processing and real refunds.
- Delivery logistics: address entry, delivery fees, rider tracking.
- Kitchen or point-of-sale integration, and staff-side table management.
- Email or SMS notifications.
- Languages other than English, and currencies other than rupees.
- Native mobile apps.
- Email verification, password reset and two-factor sign-in.

4.3 Principles that shaped the product

1. Ground every fact in the menu.
2. Maths is code, not language.
3. Consent is a button, enforced by the server.
4. Say no honestly and offer real alternatives.
5. Stay in scope; politely refuse everything else.
6. Measure changes; do not judge a prompt by feel.
7. Treat cost as a product requirement.
8. The customer controls their own data.
9. Be open in the interface: say it is an AI, and tell people to confirm allergies with the restaurant.

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5. Solution overview

Choose how you want to eat. The landing page opens with a Delivery or Dining choice. I modelled this on how Uber Eats puts an intent switch at the top of the screen, and on how Swiggy Dineout and OpenTable-style flows handle date, party size and live time slots.

- **Delivery** leads to the AI-waiter ordering flow.
- **Dining** leads to "Find a table": date, party size, time, live availability, instant confirmation.

Order with the AI waiter. Browse the 42-dish menu by course with no AI involved, or ask free-form questions. Add dishes, see the cart, read the order back, and confirm with the button.

After the order. PDF invoice, star rating, cancel within 10 minutes for a full refund (recorded, not paid out), and a bill split for 2 to 12 people, equally or dish by dish.

Allergen Shield. The customer saves "vegetarian only" and any of nuts, dairy, gluten, fish or egg to avoid. Every dish card then says whether it fits the profile, and adding a conflicting dish shows a warning.

Table booking. Date up to 30 days ahead, party of 1 to 8, 14 seatings (lunch 12:00 to 14:30, dinner 19:00 to 22:30), 90-minute hold, 11 tables and 44 seats assigned by best fit, free cancellation until two hours before, and a calendar file.

A decision I made from user feedback: food for dining can be pre-ordered only after a table is booked. Ordering food for a dining visit without a table made no sense, and the first version let people do it. Choosing Dining now shows a single "Book your table first" screen, with the menu and chat unlocked only once a booking is selected. The server enforces this too, so the page cannot be bypassed. Customers can cancel a booking from that same screen.

Your data. Download everything stored about you, or delete the account (password required). Deleting an account also frees its tables and removes the name and notes from its bookings.

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6. Requirements

Priority: P0 means the product must not ship without it, P1 is core value, P2 is a useful extra. The last column names how it is verified.

6.1 Menu and discovery

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-1	A 42-dish menu with price, diet, spice, allergens and photo	All fields present; 29 dishes vegetarian	P0	Menu checks
FR-2	Course picker that needs no AI	"Show me the menu" gives course buttons, then dishes from the data, with no model call	P0	Course checks
FR-3	Free-form questions answered through a menu lookup only	The AI cannot name a dish or price the lookup did not return; an unknown dish gets an honest no plus real alternatives	P0	Tool checks, evaluation
FR-4	Dish card with price, diet mark, spice, allergens, quantity 1 to 20, instant "Added" feedback	Visible and usable by keyboard and touch	P1	Contrast and layout checks

6.2 Cart and ordering

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-5	Cart add, remove, reduce; maximum 20 per dish; totals computed in code	Totals equal the sum of lines; the AI never states its own total	P0	Cart checks
FR-6	Add, remove, checkout and confirm work without the AI	Still work when every AI provider is rate-limited	P0	Order-flow checks
FR-7	Read back every line and the exact total, then a Yes, place order button	An order is placed only if the customer is signed in and pressed the button for the current cart; any change to the cart cancels that consent	P0	Guardrail checks
FR-8	Ordering needs an account; guests can browse and keep a cart	A guest at checkout is asked to sign in and the cart is kept	P1	Session checks
FR-9	Voice input with strict command matching	Casual words such as "okay" never place an order	P1	Voice and command checks

6.3 After the order

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-10	PDF invoice for the customer's own orders	Another customer's order returns "not found"; cancelled orders are stamped	P1	Invoice checks
FR-11	1 to 5 star rating with an optional comment (500 characters)	Own orders only; editable	P2	Feedback checks
FR-12	Cancel within 10 minutes with a full refund entry	Exactly one refund even on a double click or two tabs; the AI has no cancel tool	P0	Cancel checks
FR-13	Bill split for 2 to 12 people, equal or by dish	Shares always add up to the total, tested on 300 random splits; amounts recomputed on the server; locked once anyone is marked paid	P1	Split checks

6.4 Allergen Shield

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-14	Saved food profile: vegetarian only, avoid nuts, dairy, gluten, fish, egg	Input strictly validated	P1	Profile checks
FR-15	Dish cards show whether the dish fits the profile; adding a conflicting dish warns	All 42 dishes against all 5 allergens flag exactly as the menu data says	P0	Profile checks
FR-16	A profile note is added to the AI's instructions	Present only when a profile exists; documented as guidance, not a guarantee	P1	Profile checks
FR-17	An allergy caveat wherever allergens appear	Wording present on every such screen	P0	Review

6.5 Delivery or Dining, and table booking

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-18	Delivery or Dining choice at the start of the landing page and ordering screen	Delivery opens AI ordering; Dining opens Find a table	P1	Booking checks
FR-19	Live-slot booking with the rules in section 5	The server validates every field; past or closed slots are refused	P0	Booking checks
FR-20	No double-booking under simultaneous requests	12 simultaneous requests for the one 8-seat table give exactly one success; 20 requests for 11 tables give 11	P0	Booking checks
FR-21	One booking per overlapping time and at most 3 upcoming per customer	Enforced on the server	P1	Booking checks
FR-22	Dining food can be ordered only after a table is booked	The menu, cart and chat stay hidden until a booking is selected; the chat service returns a 403 for any dining request without one of the customer's own upcoming bookings, including forged or past booking ids	P0	Booking checks, live check
FR-23	Cancel a booking (owner only, until two hours before), from the account page and from the dining screen	Inside two hours the screen asks the customer to call; cancelling frees the table	P1	Booking checks
FR-24	After signing in, return the customer to where they came from	Only same-site booking and ordering pages are accepted; outside links are ignored	P1	Booking checks
FR-25	Calendar file for a booking, with no personal details	File opens in common calendars	P2	Booking checks
FR-26	Deleting an account frees its tables and removes name and notes from every booking	Verified after deletion	P0	Booking checks

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-32	Book, cancel and move a table, and cancel a food order, by chat or voice in plain language	"book 8 pm today there are 4 people" books a table for 4 at 8 pm; "we can't make it" cancels it; "make it 9 instead" moves it; "I changed my mind about the food" cancels the order. The AI only reads the words into fields; the fields are validated and the request goes through the same booking and cancel rules as the buttons	P0	Chat booking, plain-language and cancel checks
FR-33	Delivery and Dining each keep their own conversation, cart and voice session	Switching modes shows each mode's own history; nothing from Delivery appears under Dining	P1	Browser test

6.6 Data and trust

ID	Requirement	Acceptance criterion	Priority	Verified by
FR-27	Data export as JSON	Includes everything stored about the account, never the password hash	P1	Privacy checks
FR-28	Account deletion with password confirmation	Account removed; bookings freed and scrubbed	P0	Privacy checks
FR-29	Privacy policy and terms that match what the product does	Text reviewed against behaviour (legal review still to do)	P1	Review
FR-30	On-screen notice that answers come from an AI and allergies should be confirmed with the restaurant	Visible on the ordering screen	P0	Review
FR-31	About page crediting the creator and linking to the evidence	Page live	P2	Review

6.7 Not built yet

Real delivery flow, reminders, a waitlist, staff table management, one-tap reorder, a "next available" suggestion when a slot is full, and Hindi or other languages.

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7. How the AI works and where the guardrails sit

7.1 Design

The assistant has five tools: search the menu, add to cart, read the cart, place the order and check an order. It has **no** tool for cancelling, refunding, splitting or booking, so it cannot do those things however it is asked.

It does not see the menu in its prompt. Dishes, prices and allergens reach it only as results of the menu search, in a compact text form to save tokens. Each turn is limited to 8 tool steps, 1,200 output tokens and the last 40 messages.

Model fallback: continuity and cost. One model is a single point of failure and an open-ended bill, so the assistant runs on a chain of four models, tried in order. The first three are free: Groq gpt-oss-120b, Groq gpt-oss-20b and Google Gemini 3.5 Flash. The last is the only paid one, Anthropic Claude Haiku.

- *Why a chain.* Free tiers are small (Groq's allows roughly 200,000 tokens a day and 8,000 a minute; Gemini's about 20 requests a day), so no single free model can carry a public launch. Chaining several lets free capacity cover most traffic, and the paid model is only reached when all the free ones have failed.
- *Disruption.* If a model is rate-limited, out of quota or down, the next one takes over within the same message. Each attempt is run to the end and held back, so the customer sees one clean answer. A failed attempt is rolled back (the cart is restored) and the next model reruns the whole turn, so nothing is added twice.
- *Time limits.* A stuck model is abandoned after 12 seconds (22 for the last one), so a hung provider cannot freeze the chat.
- *Rest period.* A provider that just failed is skipped for 2 minutes, or 10 minutes for a daily-quota error, so the chain does not keep retrying a model that is known to be out.
- *Cost cap.* The paid model is skipped once the shared daily budget is used (default 60 calls a day across all visitors). Setting the cap to 0 turns it off. The budget fails closed: if the counter cannot be read, the paid model is not used.
- *If every model fails.* The customer gets a plain message and the buttons still work, because menu, cart, checkout, cancel and booking do not need a model.
- *Trade-off.* The reply arrives all at once instead of streaming word by word, and the free models are less capable than the paid one. That is why section 9 proposes scoring every model in the chain, and deciding which may answer allergen-related turns.

Steps that need no AI at all: the course picker, add and remove, checkout, confirm, cancel, split, and booking.

7.2 What the assistant must and must not do

Must	Must not
Use the menu search for every dish, price and allergen fact	State a dish, price or allergen that the search did not return
Say it does not have a dish and offer only real alternatives	Invent an alternative
Ask a short question when a request is ambiguous	Guess items or quantities
Leave all arithmetic to the tools	Work out or repeat a total of its own
Read the cart back and ask the customer to press the button	Treat a typed "yes", or its own summary, as consent
Politely refuse off-topic requests and instructions hidden in user text	Follow them
Mention the allergy caveat when allergens come up	Call a dish "allergen-free"
Keep answering when a model is rate-limited or down, by trying the next model in the chain (free ones first)	Spend on the paid model past the shared daily cap, or leave a half-finished cart after a failed attempt

7.3 Guardrails

I enforce each rule at the layer where it can be checked by code, rather than relying on the prompt.

Area	Control	Where it lives
Input	400-character message limit; 400 KB request limit; a fixed check for off-topic and injection attempts before the model sees anything; chat history treated as untrusted (text only, trimmed, forged system or tool parts removed)	Chat service
Grounding	Menu reachable only through the search tool; empty result for unknown dishes; a post-reply check that flags unknown rupee amounts and dish-like phrases	Tools, plus a logged check

Area	Control	Where it lives
Actions	Placing an order needs a signed-in customer and a recorded button press for the current cart; no tool exists for cancel, refund, split or booking	Tool layer and server
Authorisation	Every order, invoice, rating, cancel, split, booking and export checks that the item belongs to the signed-in account	API routes
Business rules	Cancel window, one refund per order, server-side split maths, booking capacity and race protection, dining needs a booking	Stores and routes
Allergens	Flags and warnings computed from menu data; the AI's wording is guided	Cards and order flow
Cost	Free models first; shared daily cap on the paid model that fails closed; rate limits per visitor and overall; compact prompts	Budget and rate-limit modules
Privacy and security	Minimal data; export and deletion; HTTPS and security headers; hashed passwords; signed HttpOnly session cookie; rate-limited sign-in	Account and configuration

7.4 What building this taught me

- My first prompt answered off-topic questions, added up a "₹600 total" on its own and tried to place an order without reading it back.
- The free models treated "place my order", even their own summary, as consent. That is why consent became a server-recorded button press.
- My first confirmation design used a separate route. On a serverless host each route has its own memory, so it saw an empty cart. Confirmation now lives in the chat service.
- A live booking failed to save because of a storage version conflict. I replaced three hand-written versions with one tested helper and added a self-check endpoint.

7.5 Known limits

The AI's free-text allergen wording is guided, not guaranteed; the dish flags and the add-to-cart warning are the enforced layer. The post-reply grounding check can flag a reply but cannot take back text that has already streamed. I have not yet run a formal red-team exercise.

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8. Success metrics

The product is not instrumented for a funnel yet, so the targets below are the ones I would hold it to once it is. They give a baseline to set first, then improve against.

8.1 North Star

Trusted orders completed per week: orders placed with no grounding violation in the conversation, with the customer's own confirmation press, and without a cancellation caused by a wrong item or total.

I chose it over order volume or chat volume because it rewards correct, consented orders. Speed bought at the cost of accuracy is exactly how the drive-thru pilots failed.

8.2 Metric tree

Stage	Metric	How it is defined	Target
Acquisition	Landing to chat start	Sessions opening the order page ÷ landing sessions	Set baseline
Acquisition	Delivery versus Dining mix	Share choosing each at the start	Track
Activation	Time to first add	Seconds from opening the order page to first dish added	Median 60 s or less
Conversion	Cart to order	Orders ÷ carts that reached checkout	Set baseline
Conversion	Sign-in drop-off	Carts abandoned at the sign-in step	Falling over time
Conversion	Booking completion	Confirmed bookings ÷ sessions with a slot chosen	Set baseline
AI quality	Hallucination rate (evaluation)	Answers with an invented dish or price ÷ graded answers	0% (currently 0.0% on 46 cases)
AI quality	Grounding flag rate (live)	Flagged turns ÷ answered turns	0.5% or less
AI quality	Correct decline rate	Off-topic and not-on-menu requests handled correctly	95% or more (currently 100%)
AI quality	Allergen flag accuracy	Dishes and allergens flagged exactly per menu data	100% (checked for all 42 × 5)
Retention	30-day repeat order	Customers with 2 or more orders in 30 days	Set baseline
Retention	Allergen Shield adoption	Signed-in customers with a saved profile	Track
Satisfaction	Average rating	1 to 5 stars	4.3 or more once there are 30 ratings
Operations	Cancel and refund rate	Cancelled ÷ placed	Watch for spikes
Operations	Share of turns with no AI call	Turns answered without a model	High and stable
Operations	p95 reply time	Request to first reply	6 s or less
Cost	Paid-model calls per day	Against the shared cap	At or under the cap (default 60)
Cost and continuity	Free-model share of AI answers	Answers given by a free model ÷ all AI answers	95% or more
Continuity	Turns answered after a fallback	Turns where the first model failed and a later one answered ÷ all AI turns	Track; the customer-facing failure rate should stay near 0%
Cost	AI cost per completed order	Model spend ÷ orders	1% or less of average order value

8.3 Guardrail metrics

A release is blocked if any of these moves the wrong way: an invented dish or price on the evaluation set, any order without a recorded confirmation, any cross-account data read, allergen flag accuracy below 100%, split shares not equal to the total, a double refund, a double-booked table, or the paid cap exceeded.

8.4 Counter-metrics

If this improves	Check that this has not got worse
Conversion	Refund and cancel rate, complaints
Share of turns with no AI	Unresolved-question rate
Reply speed	Grounding flag rate
Allergen warnings shown	Add-to-cart rate on flagged dishes (warning fatigue)

8.5 Instrumentation plan

Events with no personal data: landing view, service selected, chat started, course picked, AI answer (model, paid or not, tokens, steps, grounding flag), add to cart, checkout started, sign-in prompted, order confirmed, order placed, order cancelled, rating submitted, profile saved, booking searched, confirmed and cancelled. Today the server logs only usage, failover, grounding and session lines, and the host keeps about an hour of them.

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9. Evaluating the AI

9.1 Approach

I evaluate the whole system (prompt, tools, guardrails, failover), not only the model. I separate capability tests (does it help?) from risk tests (can it hurt?). Risk tests are hard gates: if one fails, the release does not go out.

9.2 Current results

The evaluation set has 46 labelled cases across menu facts, dietary needs, cart, honest-no, off-topic, ambiguous requests and confirmation. I ran it once at build time on Claude Sonnet 5.5.

Metric	First prompt	Tuned prompt
Tool selection accuracy	96.7%	100%
Argument accuracy	100%	100%
Grounding rate	95.7%	100%
Hallucination rate	4.3%	0.0%
Correct decline (off-topic and not-on-menu)	84.6%	100%
Cart and total correctness	100%	100%
Order safety	66.7%	100%
Overall case pass	87.0%	100%

The evaluation earned its keep: it caught the model reading the cart back from memory (the guard rightly refused the order) and a read-back plus order in a single turn that slipped through an early guard.

9.3 How far to trust these numbers

Not very far, yet. It was one run, with only 46 cases, written by me and kept in the same repository as the product, so it was not held out. It ran on a different model from the one that mostly serves live traffic (the free models),

and it does not score tone or helpfulness. It is English only and has no hard allergen cases. The 100% figures show that the guardrails work on these cases, not that production accuracy is 100%.

9.4 The evaluation suite I would build next

Suite	What it tests	Cases	How it is graded	Gate
A. Facts and grounding	Dish, price, spice and allergen questions	80	Automatically against menu data	No invented facts
B. Honest no	Dishes not on the menu, including near-misses such as "paneer tikka masala"	40	Automatic plus rubric	95% or more
C. Allergen and diet	"Nut-free", "vegan", "gluten-free", combined constraints, misspellings, pressure to confirm safety	60	Automatic; any "allergen-free" claim fails	Zero unsafe claims
D. Cart and maths	Quantities, removals, totals	40	Automatic	100%
E. Consent and actions	"Just place it", typed yes, role-play, instructions hidden in history	40	Automatic on the tool calls made	Zero orders without the button
F. Scope and injection	Off-topic, jailbreaks, instructions hidden in notes or dish names	60	Rubric plus automatic	98% or more safe
G. Ambiguity	Under-specified requests	30	AI judge, sampled by a human	85% or more clarified
H. Tone and helpfulness	Brevity, politeness, usefulness	40	AI judge calibrated to human labels	No regression
I. Hinglish and Hindi	Real user phrasing	40	Mixed	Set baseline

That is about 430 cases, with 20% held out and never used to tune a prompt.

9.5 Method

- Score every model in the live chain, not only the best one, and decide which models may answer allergen-related turns.
- Repeat each run three to five times and report the spread, so flaky cases are visible.
- Use an AI judge only for the subjective suites (G and H), after checking it against at least 100 human-labelled examples and reporting how often it agrees with them.
- Re-run suites A to F whenever the prompt, the model chain or a tool changes. The hard gates run in the build pipeline and the results are kept.
- Run the paid model in evaluations only at release time, within a set budget.

9.6 Red-team session

One structured session covering prompt injection through every field a user controls, social engineering for refunds or orders, pressure on allergen answers ("you said it was safe"), forged history, attempts to burn through the AI budget, and attempts to read another customer's data. Findings are logged by severity and added to the evaluation set.

9.7 Monitoring after launch

Signal	Source	Action
Grounding flags ÷ answered turns	Grounding log	Above 0.5%: review a sample, tighten the prompt
Failover rate	Failover log	Rising: provider problem
Paid calls versus cap	Usage log	Above 80% of cap: alert
Cancel and refund spikes	Order events	Check for wrong-item causes
Complaints and thumbs-down (to be added)	Feedback	Weekly review of 50 random and all flagged
Conversations that mention allergens	Chat log	Human review of every one during any pilot

9.8 Release gates

To ship, all of these must hold: no invented facts on suites A and B, no unsafe allergen claims on C, no orders without the button on E, cart maths 100%, all automated checks passing (759 passing across 25 suites today), no new serious red-team finding, and the paid cap verified. Correct decline of 95% or more, tone not regressing and the reply-time target are tracked as softer gates.

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10. Non-functional requirements

Area	Requirement	Where it stands
Accessibility	Target WCAG 2.2 AA. An automated contrast test reads the real theme colours; it found two failures (white on the brand red and orange) and both were fixed	Automated checks and emulated screen sizes only. A screen-reader test is still to do
Security	HTTPS, strict headers, hashed passwords, signed session cookie, rate limits, ownership checks	Suitable for a demonstration. Email verification, password reset and two-factor are not built
Privacy	Stores name, email, password hash, orders, splits, ratings, food profile and bookings. No addresses or payment details. Two essential cookies only	Policy text has not had legal review
Performance	Typical page about 0.3 s from a static landing page	About 1 in 100 requests stalls for roughly 20 s at the hosting layer. I measured this from one network. A retry prompt appears after 6 s
Reliability	The chat always answers because deterministic paths do not depend on the AI. Storage errors do not crash pages	No alerting yet
Concurrency	One tested conditional-write helper for accounts, bookings and the daily budget	A production self-check endpoint passes
Cost	At default limits, worst-case paid spend is a few tenths of a dollar a day; setting the paid cap to 0 turns the paid model off	I still need to set a monthly spend limit in the Anthropic console

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11. Launch and rollout

- **Done:** build, automated checks, a preview deployment for every feature, and my approval before every production release. Rollback means redeploying the previous build.

- **Portfolio launch:** share the live product on LinkedIn. Before posting I will set the Anthropic spend limit, check the share preview, and do a final run as a stranger: sign up, book a table, cancel it, place an order and rate it.
- **Pilot with a real restaurant (a later decision):** only after evaluation suites A to F pass on every model in the chain, allergen conversations are reviewed by a human, and the terms and privacy text have been reviewed by a lawyer.
- **Controls if something goes wrong:** set the paid cap to 0, lower the per-visitor limits, or turn on the host's attack-challenge mode.

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12. Risks, open questions and next steps

12.1 Risks

Risk	Likelihood	Impact	Mitigation	What remains
The AI states a wrong allergen fact	Medium	High	Flags in code, caveat everywhere, allergen gate per model, human review in a pilot	AI wording is guided, not guaranteed
A traffic spike drains paid credits	Medium	Medium	Shared cap, free models first, rate limits, spend limit	Low
Free models slow down or hit limits	High	Medium	Failover and no-AI paths	Replies may be slower
Customers mistake the product for a live restaurant	Medium	Medium	Clear notes in the interface and policies	Low
Storage conflicts or lost carts	Medium	Medium	Conditional-write helper, self-check; carts in server memory can reset	Carts may reset
Occasional 20-second stalls	Medium	Low to medium	Retry prompt; review hosting region	Not solved
Prompt injection	Medium	Medium	Input check, sanitised history, no action tools for the AI	Needs the red-team session
Privacy or legal exposure	Low	Medium	Minimal data, export and deletion	Needs legal review

12.2 What I still need to learn

I have not tested the core assumptions with people: that diners want to order conversationally from a single restaurant, that operators would adopt it, that a 10-minute cancel window and a two-hour booking cut-off match how a real kitchen works, and that voice and group ordering will actually be used.

First four weeks:

1. Interview 5 to 8 diners, especially people with allergies, and 2 to 3 independent restaurant owners.
2. Watch a small group complete a booking and an order, without helping.
3. Add the funnel events in section 8.5 and set the baselines.
4. Build evaluation suites A to F and run them on every model in the chain.
5. Run one red-team session and fix what it finds.

12.3 Open questions

- Is the main audience restaurant operators or reviewers of the portfolio? The About page currently serves both.
- Should the Allergen Shield block a conflicting dish rather than only warn?
- Should "paid" tracking in a split connect to a real UPI link later?
- Is a partial refund needed after the 10-minute window?
- Should I buy a custom domain before the launch post?
- Is a different hosting region the fix for the stalls?

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13. Roadmap

Order	Item	Why	Size
1	Expand the evaluation set, score every model, gate releases	Biggest gap in safety and credibility	Medium
2	Add funnel instrumentation	No usage data today	Small
3	User interviews and an observed pilot	Validate the problem	Small
4	Email verification and password reset	Account safety	Medium
5	Real delivery flow: address, fee, ETA, tracking	Biggest functional gap	Large
6	Booking reminders and a "next available" suggestion	Fewer no-shows	Medium
7	A real database for accounts, carts and orders	Durability	Large
8	Payments sandbox with real refunds	Completes the order loop	Large
9	Hindi and Hinglish with matching evaluations	Wider reach	Medium
10	Screen-reader pass and legal review	Compliance	Small to medium

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Appendix A. Technical summary

Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS 4, Zod 4 and the Vercel AI SDK 7, with pdf-lib for invoices and Vercel Blob for storage. Hosted on Vercel: a push to the main branch deploys to production and other branches build previews.

Accounts, bookings and the daily budget counter live in a private Blob store behind one conditional-write helper. Carts live in server memory. Pages: home, order, book, account, sign in, sign up, about, privacy and terms. APIs cover chat, feedback, invoices, order cancel and split, profile, data export, bookings (availability, create, cancel) and health checks.

Appendix B. Data model

- **Account:** id, name, email, password hash, created date, profile, orders, bookings.
- **Order:** id, total, time placed, lines, service (delivery or dine-in with an optional booking id), feedback, cancellation and refund details, split.

- **Booking:** id, customer, name, date, time, party size, table, notes, status. Stored one file per day.

Appendix C. References

No.	Topic	Source	Link
1	Swiggy ordering through ChatGPT, Claude and Gemini (January 2026)	YourStory	yourstory.com/2026/01/swiggy-lets-users-order-through-chatgpt-claude-gemini
2	Zomato's in-app AI assistant	Inc42	inc42.com/buzz/zomato-introduces-ai-powered-assistant-to-help-customers-place-orders
3	McDonald's ending its AI drive-thru test (June 2024)	Information Age (ACS)	ia.acs.org.au/article/2024/mcdonald-s-bins-ai-drive-thru-after-errors-go-viral.html
4	Guidance on allergen answers in restaurant AI	Hostie	hostie.ai/resources/voice-ai-allergy-dietary-questions-menu-aware-bots-2025
5	Structure for AI product requirement documents	Uvik	uvik.net/blog/ai-prd-template

If a link does not open from your PDF viewer, copy the address shown in the Link column into a browser.